

Extra Practice Questions

1 Intuitive Method

Use the intuitive method to show how to get these (regular) demand functions. (Hint for the 2nd example, which is a bit tricky on corner solutions: first explain why the $x_1 = 0$ corner is never optimal, then explain why if $\frac{p_1}{p_2} < 2$, then good 1 ALWAYS has more bang-per-buck. Finally, find the interior optimal solution if $\frac{p_1}{p_2} > 2$).

utility function	demand
$\min\{x_1, 2x_2\}$	$\left(\frac{2m}{2p_1+p_2}, \frac{2m}{2p_1+p_2}\right)$
$x_1^2 + x_1x_2$	$\begin{cases} \left(\frac{m}{p_1}, 0\right) & \text{if } p_1 < 2p_2 \\ \left(\frac{m}{2(p_1-p_2)}, \left(\frac{p_1}{p_2} - 2\right) \frac{m}{2(p_1-p_2)}\right) & \text{if } p_1 > 2p_2 \end{cases}$
$\ln x_1^2 + \ln x_2$	$\left(\frac{2}{3} \frac{m}{p_1}, \frac{1}{3} \frac{m}{p_2}\right)$
$2x_1 + x_2$	$\begin{cases} \left(\frac{m}{p_1}, 0\right) & \text{if } p_1 < 2p_2 \\ \left(0, \frac{m}{p_2}\right) & \text{if } p_1 > 2p_2 \end{cases}$
$2x_1^2 + 2x_2^2$	$\begin{cases} \left(\frac{m}{p_1}, 0\right) & \text{if } p_1 < p_2 \\ \left(0, \frac{m}{p_2}\right) & \text{if } p_1 > p_2 \end{cases}$

1. $u(x_1, x_2) = \min\{x_1, 2x_2\}$

Answer: we want the best affordable bundle. First, allocating money smartly between goods here entails equating the expressions in square brackets (these are perfect complements, and not wasting any money means not buying extra of anything, it uses money without providing any utility). Second, we want to spend all money. So solve $x_1 = 2x_2$ and $p_1x_1 + p_2x_2 = m$ to get the optimal bundle indicated above.

2. (Harder) $u(x_1, x_2) = x_1^2 + x_1x_2$

Following the hint, first note that the $x_1 = 0$ corner gives utility zero; clearly this is worse than the $x_2 = 0$ corner (which gives utility $\left(\frac{m}{p_1}\right)^2$) and thus is never optimal. To see why the all-\$-on-good-1 corner is optimal when $p_1 < 2p_2$, take derivatives to obtain

$$\begin{aligned} MU_1 &= 2x_1 + x_2 \text{ and } MU_2 = x_1 \\ \Rightarrow \frac{MU_1}{MU_2} &= \frac{2x_1 + x_2}{x_1} = 2 + \frac{x_2}{x_1} \end{aligned}$$

By nonnegativity of x_1 and x_2 , this is always at least 2. So if $\frac{p_1}{p_2} < 2$, then any feasible bundle has $\frac{MU_1}{MU_2} > \frac{p_1}{p_2}$, i.e. good 1 always has more bang-per-buck, and thus the optimal solution in this case is the corner $\left(\frac{m}{p_1}, 0\right)$.

Finally suppose $p_1 > 2p_2$. Notice that these preferences are convex, since the MRS $2 + \frac{x_2}{x_1}$ falls as $x_1 \uparrow$ and $x_2 \downarrow$ (good 1's relative value falls as it becomes relatively more abundant).

So we're looking for an interior bundle unless forced into a corner,, and thus proceed with the usual method: equating bang-per-buck,

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2} \Rightarrow 2 + \frac{x_2}{x_1} = \frac{p_1}{p_2} \Rightarrow x_2 = \left(\frac{p_1}{p_2} - 2 \right) x_1$$

And then plugging this into the budget line,

$$p_1 x_1 + p_2 x_2 = m \Rightarrow p_1 x_1 + p_2 \left(\frac{p_1}{p_2} - 2 \right) x_1 = m \Rightarrow x_1 = \frac{m}{2(p_1 - p_2)}$$

And then plugging this back into the equal-bang-per-buck condition, $x_2 = \left(\frac{p_1}{p_2} - 2 \right) \frac{m}{2(p_1 - p_2)}$. Now, as long as these are both positive, this is the solution: but this is indeed the case for the range under consideration, $p_1 > 2p_2$. Summing up, if $p_1 < 2p_2$ then the previous paragraph explains why you should spend all \$ on good 1, so demand is $\left(\frac{m}{p_1}, 0 \right)$; if $p_1 > 2p_2$ then this paragraph finds the optimal interior bundle. At $p_1 = 2p_2$, either bundle is optimal.

3. $u(x_1, x_2) = \ln x_1^2 + \ln x_2$.

Shortcut option: these are equivalent to $\ln x_1^2 x_2$ which is equivalent to $x_1^{\frac{2}{3}} x_2^{\frac{1}{3}}$, so spend $\frac{2}{3}$ income on good 1 and $\frac{1}{3}$ on good 2 by the magic rule, i.e.

$$x_1 = \frac{2}{3} \frac{m}{p_1} \text{ and } x_2 = \frac{1}{3} \frac{m}{p_2}$$

Full intuitive method: here since $\ln x_1^2 = 2 \ln x_1$, we have $MU_1 = \frac{2}{x_1}$ and $MU_2 = \frac{1}{x_2}$. So $\frac{MU_1}{MU_2} = 2 \frac{x_2}{x_1}$; thus falls as $x_1 \uparrow$ and $x_2 \downarrow$, establishing convexity. Equating bang-per-buck,

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2} \Rightarrow \frac{2/x_1}{1/x_2} = \frac{p_1}{p_2} \Rightarrow 2 \frac{x_2}{x_1} = \frac{p_1}{p_2} \Rightarrow p_1 x_1 = 2 p_2 x_2$$

Plugging this into the budget line $p_1 x_1 + p_2 x_2 = m$, we get

$$2 p_2 x_2 + p_2 x_2 = m \Rightarrow 3 p_2 x_2 = m \Rightarrow x_2 = \frac{1}{3} \frac{m}{p_2}$$

And plugging this back into the bang-per-buck expression, we get $x_1 = \frac{2}{3} \frac{m}{p_1}$.

4. $u(x_1, x_2) = 2x_1 + x_2$

Answer: these are perfect substitutes where good 1 has twice the value as good 2 ($\frac{MU_1}{MU_2} = 2$ at all bundles). So if good 1 has less than twice the cost of good 2, it's the better use of money, so spend all money on good 1. (Mathematically, if $\frac{p_1}{p_2} < 2$, then at any bundle we have $\frac{MU_1}{MU_2} > \frac{p_1}{p_2} \Leftrightarrow \frac{MU_1}{p_1} > \frac{MU_2}{p_2}$: good 1 has more bang-per-buck, always, so spend all money on it). If good 1 has more than twice the cost, it's a worse use of money, so spend all money on good 2. And if $\frac{p_1}{p_2} = 2$, goods 1 and 2 are equally good uses of money, so any point on the budget line is optimal. Summing up:

$$(x_1, x_2) = \begin{cases} \left(\frac{m}{p_1}, 0 \right) & \text{if } p_1 < 2p_2 \\ \left(0, \frac{m}{p_2} \right) & \text{if } p_1 > 2p_2 \end{cases}$$

(And at $p_1 = 2p_2$, any bundle with $x_1 \in [0, \frac{m}{p_1}]$ units of good 1 and $\frac{m - p_1 x_1}{p_2}$ units of good 2 is optimal).

5. $u(x_1, x_2) = 2x_1^2 + 2x_2^2$.

Answer: here $MU_1 = 4x_1$ and $MU_2 = 4x_2$, so $\frac{MU_1}{MU_2} = \frac{x_1}{x_2}$: this *rises* as $x_1 \uparrow$ and $x_2 \downarrow$, so these are concave (addictive) not convex. Go for a corner. All money on good 1 (i.e. the bundle $(\frac{m}{p_1}, 0)$) yields utility $2\left(\frac{m}{p_1}\right)^2$, and all money on good 2 yields utility $2\left(\frac{m}{p_2}\right)^2$. The first is better whenever $\frac{2m^2}{p_1^2} > \frac{2m^2}{p_2^2} \Leftrightarrow p_2 > p_1$. So the optimal bundle is the all-\$-on-good-1 corner if $p_1 < p_2$, and the all-\$-on-good-2 corner if $p_2 > p_1$ (they're equally good at $p_1 = p_2$).

2 Lagrangians

Redo the Section 1 examples with Lagrangians, if possible.

1. Not possible: we can't do Lagrangians when we can't take derivatives. Use your intuition for perfect complements.
2. The Lagrangian is

$$\mathcal{L} = x_1^2 + x_1x_2 + \lambda_1x_1 + \lambda_2x_2 + \lambda_3(m - p_1x_1 - p_2x_2)$$

The equations we definitely must satisfy are $\frac{\partial \mathcal{L}}{\partial x_1} = 0$, $\frac{\partial \mathcal{L}}{\partial x_2} = 0$, and binding budget constraint. Writing these out explicitly:

$$2x_1 + x_2 + \lambda_1 - \lambda_3p_1 = 0 \tag{1}$$

$$x_1 + \lambda_2 - \lambda_3p_2 = 0 \tag{2}$$

$$p_1x_1 + p_2x_2 = m \text{ (and } \lambda_3 > 0) \tag{3}$$

And for the nonnegativity constraints, each may in principle bind or not bind.

- Let's start looking for a solution where neither binds, so

$$\lambda_1 = 0 \text{ and } x_1 \geq 0 \tag{4}$$

$$\lambda_2 = 0 \text{ and } x_2 \geq 0 \tag{5}$$

Then, plugging (4) and (5) namely $\lambda_1 = \lambda_2 = 0$ into (1) and (2), we get

$$2x_1 + x_2 = \lambda_3p_1$$

$$x_1 = \lambda_3p_2$$

Dividing the first by the 2nd, which is safe since we have $\lambda_3 > 0$ from (3), we get

$$\frac{2x_1 + x_2}{x_1} = \frac{p_1}{p_2} \Rightarrow 2p_2x_1 + p_2x_2 = p_1x_1 \Rightarrow p_2x_2 = (p_1 - 2p_2)x_1 \tag{*}$$

Plugging this into the binding budget constraint (3),

$$p_1x_1 + (p_1 - 2p_2)x_1 = m \Rightarrow x_1 = \frac{m}{2(p_1 - p_2)}$$

Then plugging this back into (*), $x_2 = \frac{p_1 - 2p_2}{p_2} \frac{m}{2(p_1 - p_2)}$. Finally, these values work ie satisfy (4) and (5) as long as they are nonnegative, which happens if $p_1 \geq 2p_2$. And plugging into (2) we can get the final value λ_3 :

$$\lambda_3 = \frac{x_1 + \lambda_2}{p_2} = \frac{\frac{m}{2(p_1 - p_2)} + 0}{p_2} = \frac{m}{2p_2(p_1 - p_2)}$$

Summary so far: we get a solution where neither nonnegativity constraint binds whenever $p_1 \geq 2p_2$. The solution is $(x_1, x_2, \lambda_1, \lambda_2, \lambda_3) = \left(\frac{m}{2(p_1 - p_2)}, \frac{p_1 - 2p_2}{p_2} \frac{m}{2(p_1 - p_2)}, 0, 0, \frac{m}{2p_2(p_1 - p_2)} \right)$. But if $p_1 < 2p_2$, the above solution violated the nonnegativity constraint for x_2 , so we turn to this case next.

- Now assume $p_1 < 2p_2$. We saw that when we guessed non-binding nonnegativity constraints, we violated the constraint $x_2 \geq 0$, so let's guess that this one in fact does bind: let's look for a solution satisfying (1)-(4) above, but replacing (5) with its binding counterpart:

$$x_2 = 0 \text{ and } \lambda_2 > 0 \quad (5a)$$

Plugging $x_2 = 0$ into binding budget constraint (3), we get $x_1 = \frac{m}{p_1}$. So far we have $x_1 = \frac{m}{p_1}, x_2 = 0, \lambda_1 = 0$. Plug these into (1) to get

$$2 \left(\frac{m}{p_1} \right) + 0 + 0 - \lambda_3 p_1 = 0 \Rightarrow \lambda_3 = \frac{2m}{p_1^2}$$

Then plug this into (2) to get

$$\frac{m}{p_1} + \lambda_2 - \left(\frac{2m}{p_1^2} \right) p_2 = 0 \Rightarrow \lambda_2 = \frac{m}{p_1^2} (2p_2 - p_1)$$

This solves (5a) so long as it's positive, i.e. $2p_2 > p_1$, which is indeed what we assumed. So we have the solution: if $p_1 < 2p_2$, then $(x_1, x_2, \lambda_1, \lambda_2, \lambda_3) = \left(\frac{m}{p_1}, 0, 0, \frac{m}{p_1^2} (2p_2 - p_1), \frac{2m}{p_1^2} \right)$.

3. The Lagrangian is

$$\mathcal{L} = \ln x_1^2 + \ln x_2 + \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 (m - p_1 x_1 - p_2 x_2)$$

The conditions we definitely must satisfy are $\frac{\partial \mathcal{L}}{\partial x_1} = 0, \frac{\partial \mathcal{L}}{\partial x_2} = 0$, and the binding budget constraint: writing these out explicitly,

$$\frac{2}{x_1} + \lambda_1 - \lambda_3 p_1 = 0 \quad (6)$$

$$\frac{1}{x_2} + \lambda_2 - \lambda_3 p_2 = 0 \quad (7)$$

$$p_1 x_1 + p_2 x_2 = m \text{ and } \lambda_3 > 0 \quad (8)$$

In principle, nonnegativity constraints can bind or not bind, but we guess here that both are non-binding since buying ≤ 0 of either good gives utility $-\infty$. So the non-binding versions of the constraints are:

$$\lambda_1 = 0 \text{ and } x_1 \geq 0 \quad (9)$$

$$\lambda_2 = 0 \text{ and } x_2 \geq 0 \quad (10)$$

Now, plugging (9) and (10) namely $\lambda_1 = \lambda_2 = 0$ into (6) and (7), they become

$$\frac{2}{x_1} = \lambda_3 p_1 \text{ (6a) and } \frac{1}{x_2} = \lambda_3 p_2 \text{ (7a)}$$

Dividing (6a) by (7a), which is safe since $\lambda_3 > 0$ by (8),

$$\frac{2/x_1}{1/x_2} = \frac{p_1}{p_2} \Rightarrow 2p_2 x_2 = p_1 x_1$$

Then plugging this into binding budget constraint (8),

$$2p_2 x_2 + p_2 x_2 = m \Rightarrow x_2 = \frac{1}{3} \frac{m}{p_2}$$

And plugging this back into the previous equation, $p_1 x_1 = 2p_2 x_2 \Rightarrow x_1 = \frac{2p_2}{p_1} \left(\frac{1}{3} \frac{m}{p_2} \right) = \frac{2}{3} \frac{m}{p_1}$.

So far we have $(x_1, x_2, \lambda_1, \lambda_2) = \left(\frac{2}{3} \frac{m}{p_1}, \frac{1}{3} \frac{m}{p_2}, 0, 0 \right)$. Finally plug these into (1) or (2) to get $\lambda_3 = \frac{3}{m}$. This satisfies all constraints so is the solution.

4. The Lagrangian is

$$\mathcal{L} = 2x_1 + x_2 + \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 (m - p_1 x_1 - p_2 x_2)$$

As usual, we need to satisfy $\frac{\partial \mathcal{L}}{\partial x_1} = 0$, $\frac{\partial \mathcal{L}}{\partial x_2} = 0$, and the binding budget constraint. These are:

$$2 + \lambda_1 - \lambda_3 p_1 = 0 \tag{11}$$

$$1 + \lambda_2 - \lambda_3 p_2 = 0 \tag{12}$$

$$p_1 x_1 + p_2 x_2 = m \text{ and } \lambda_3 > 0 \tag{13}$$

We have a guess from above that if $\frac{p_1}{p_2} < 2$, then good 1 is the better use of money so the constraint $x_2 \geq 0$ binds; and if $\frac{p_1}{p_2} > 2$, then the constraint $x_1 \geq 0$ binds. But I'll do this without a guess to demonstrate how it works.

- let's first look for a solution where neither nonnegativity constraint binds. Then in addition to (11)-(13) above, we want to solve

$$x_1 \geq 0 \text{ and } \lambda_1 = 0 \tag{14}$$

$$x_2 \geq 0 \text{ and } \lambda_2 = 0 \tag{15}$$

Plugging (14) and (15) namely $\lambda_1 = \lambda_2 = 0$ into (11) and (12), they become

$$2 = \lambda_3 p_1 \text{ (11a) and } 1 = \lambda_3 p_2 \text{ (12a)}$$

So $\lambda_3 = \frac{2}{p_1}$ and $\lambda_3 = \frac{1}{p_2}$: if these happen to be the same, namely if $\frac{p_1}{p_2} = 2$, then we have a solution here. It is $\lambda_1 = \lambda_2 = 0, \lambda_3 = \frac{2}{p_1} = \frac{1}{p_2}$, and then any bundle (x_1, x_2) on the budget line $p_1 x_1 + p_2 x_2 = m$ is optimal. But if $\frac{p_1}{p_2} \neq 2$, then we have contradicting values for λ_3 so no solution here; we move on to consider the cases where nonnegativity binds for one of the goods.

- let's next look for a solution where nonnegativity binds for good 1. So we want to solve (11)-(13) and (15) from above, but replace (14) by its binding counterpart (14a): $x_1 = 0$ and $\lambda_1 > 0$. Plugging $x_1 = 0$ into the binding budget constraint (13), we get $x_2 = \frac{m}{p_2}$. Plugging (15) namely $\lambda_2 = 0$ into (12),

$$1 + 0 - \lambda_3 p_2 = 0 \Rightarrow \lambda_3 = \frac{1}{p_2}$$

Plugging this into (11) we get

$$\lambda_1 = \lambda_3 p_1 - 2 = \frac{p_1}{p_2} - 2$$

This satisfies (14a) provided that $\frac{p_1}{p_2} > 2$, so we have the solution in this case: $(x_1, x_2, \lambda_1, \lambda_2, \lambda_3) = \left(0, \frac{m}{p_2}, \frac{p_1}{p_2} - 2, 0, \frac{1}{p_2}\right)$

- and finally, let's look for a solution where nonnegativity binds for good 2. So we want to solve (11)-(14) from above but replace (15) by its binding counterpart (15a): $x_2 = 0$ and $\lambda_2 > 0$. Plugging $x_2 = 0$ into the binding budget constraint (13), we get $x_1 = \frac{m}{p_1}$. Plugging this along with $\lambda_1 = 0$ into (11), we get $2 + 0 - \lambda_3 p_1 = 0 \Rightarrow \lambda_3 = \frac{2}{p_1}$. Finally, this into (12) gives

$$\lambda_2 = \lambda_3 p_2 - 1 = 2 \frac{p_2}{p_1} - 1$$

This solves (15a) so long as it's positive, namely $2 \frac{p_2}{p_1} > 1 \Leftrightarrow \frac{p_1}{p_2} < 2$, so we have the solution in this case: it is $(x_1, x_2, \lambda_1, \lambda_2, \lambda_3) = \left(\frac{m}{p_1}, 0, 0, 2 \frac{p_2}{p_1} - 1, \frac{2}{p_1}\right)$.

- Summing up, the middle bullet point (nonnegativity binding for good 1) applies if $\frac{p_1}{p_2} > 2$, and in this case demand is $(x_1, x_2) = \left(0, \frac{m}{p_2}\right)$. The last bullet point (nonnegativity binding for good 2) applies if $\frac{p_1}{p_2} < 2$, and in this case demand is $(x_1, x_2) = \left(\frac{m}{p_1}, 0\right)$. And the first bullet point (nonnegativity constraints non binding) applies if $\frac{p_1}{p_2} = 2$, and in this case any point on the budget line is optimal.

5. Does not apply.. These are concave preferences, so jump directly to a corner solution. Lagrangians only give the right answer for convex preferences.

3 Compensated Demand

Using the intuitive method, find the compensated demand functions and expenditure functions in examples (1)-(5) above. Namely, show how to get the following answers:

utility	compensated demand	expenditure
$\min\{x_1, 2x_2\}$	$(u^*, \frac{1}{2}u^*)$	$p_1u^* + \frac{1}{2}p_2u^*$
$x_1^2 + x_1x_2$	$\begin{cases} (\sqrt{u^*}, 0) & \text{if } p_1 < 2p_2 \\ \left(\sqrt{\frac{p_2u^*}{p_1-p_2}}, \left(\frac{p_1-2p_2}{p_2}\right)\sqrt{\frac{p_2u^*}{p_1-p_2}}\right) & \text{if } p_1 > 2p_2 \end{cases}$	$\begin{cases} p_1\sqrt{u^*} & p_1 < 2p_2 \\ 2\sqrt{p_2(p_1-p_2)u^*} & p_1 > 2p_2 \end{cases}$
$\ln x_1^2 + \ln x_2$	$\left(\left(\frac{2p_2}{p_1}\right)^{\frac{1}{3}} e^{\frac{u^*}{3}}, \left(\frac{p_1}{2p_2}\right)^{\frac{2}{3}} e^{\frac{u^*}{3}}\right)$	$p_1^{\frac{2}{3}}p_2^{\frac{1}{3}}e^{\frac{u^*}{3}}\left(2^{\frac{1}{3}} + \left(\frac{1}{2}\right)^{\frac{2}{3}}\right)$
$2x_1 + x_2$	$\begin{cases} (\frac{u^*}{2}, 0) & \text{if } p_1 < 2p_2 \\ (0, u^*) & \text{if } p_1 > 2p_2 \end{cases}$	$\min\left\{\frac{p_1}{2}, p_2\right\}u^*$
$2x_1^2 + 2x_2^2$	$\begin{cases} \left(\sqrt{\frac{u^*}{2}}, 0\right) & \text{if } p_1 < p_2 \\ \left(0, \sqrt{\frac{u^*}{2}}\right) & \text{if } p_1 > p_2 \end{cases}$	$\min\{p_1, p_2\} \cdot \sqrt{\frac{u^*}{2}}$

1. $u(x_1, x_2) = \min\{x_1, 2x_2\}$. For compensated demand, we want to allocate money optimally between goods, here $x_1 = 2x_2$ (equate the things in brackets so nothing is wasted), and we want a bundle that actually gives utility u^* : $u(x_1, x_2) = u^*$. So we want to solve these two equations for x_1 and x_2 . But by the optimal spending condition, $u(x_1, x_2) = x_1 = 2x_2$, so the utility constraint becomes simply $x_1 = u^*$ and $2x_2 = u^* \Rightarrow x_2 = \frac{1}{2}u^*$. This gives the compensated demand function in the chart, and the expenditure function is its cost, $p_1x_1^c + p_2x_2^c$.
2. $u(x_1, x_2) = x_1(x_1 + x_2)$. For compensated demand, first we want to allocate money optimally between goods. This is the same condition as in regular demand above, where we already showed that (i) if $\frac{p_1}{p_2} \geq 2$, then look for an interior solution where the two goods have equal bang-per-buck; and (ii) if $\frac{p_1}{p_2} < 2$, then good 1 always has more bang per buck, so buy good 1 only. But the second constraint is now the utility constraint, $x_1(x_1 + x_2) = u^*$.

So in case (i), $\frac{p_1}{p_2} > 2$, equating bang-per-buck yields

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2} \Rightarrow 2 + \frac{x_2}{x_1} = \frac{p_1}{p_2} \Rightarrow x_2 = \left(\frac{p_1}{p_2} - 2\right)x_1$$

And then plug this into the utility constraint:

$$\begin{aligned} x_1(x_1 + x_2) &= u^* \Rightarrow x_1 \left(x_1 + \left(\frac{p_1}{p_2} - 2\right)x_1\right) = u^* \Rightarrow x_1^2 \left(\frac{p_1}{p_2} - 1\right) = u^* \\ \Rightarrow x_1 &= \sqrt{\frac{u^*}{\left(\frac{p_1}{p_2} - 1\right)}} = \sqrt{\frac{p_2u^*}{p_1 - p_2}} \end{aligned}$$

And then plug this back into the above equation for x_2 ,

$$x_2 = \left(\frac{p_1 - 2p_2}{p_2}\right) \sqrt{\frac{p_2u^*}{p_1 - p_2}} = \left(\frac{p_1 - 2p_2}{p_2}\right) \sqrt{\frac{p_2u^*}{p_1 - p_2}}$$

These values of x_1 and x_2 are the compensated demand functions for the case $p_1 < 2p_2$. And the expenditure function is the cost:

$$\begin{aligned} p_1 \sqrt{\frac{p_2 u^*}{p_1 - p_2}} + p_2 \left(\frac{p_1 - 2p_2}{p_2} \right) \sqrt{\frac{p_2 u^*}{p_1 - p_2}} &= \sqrt{\frac{p_2 u^*}{p_1 - p_2}} \left(p_1 + p_2 \frac{p_1 - 2p_2}{p_2} \right) \\ &= 2(p_1 - p_2) \sqrt{\frac{p_2 u^*}{p_1 - p_2}} \end{aligned}$$

And in case (ii), $\frac{p_1}{p_2} < 2$, it is optimal to buy good 1 only. So $x_2 = 0$. Plug this into the utility constraint to get

$$x_1(x_1 + 0) = u^* \Rightarrow x_1 = \sqrt{u^*}$$

So compensated demand is $(\sqrt{u^*}, 0)$, and expenditure is its cost, $p_1 x_1^c + p_2 x_2^c = p_1 \sqrt{u^*} + p_2 \cdot 0$

3. $u(x_1, x_2) = \ln x_1^2 + \ln x_2$. Here, we already saw these are convex. So the optimal spending condition is

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2} \Rightarrow \frac{2/x_1}{1/x_2} = \frac{p_1}{p_2} \Rightarrow 2p_2 x_2 = p_1 x_1$$

And the utility constraint is

$$u^* = \ln x_1^2 + \ln x_2 = \ln x_1^2 x_2 \Rightarrow e^{u^*} = x_1^2 x_2$$

Plugging the optimal spending condition namely $x_2 = \frac{p_1}{2p_2} x_1$ into the utility constraint,

$$\begin{aligned} e^{u^*} &= x_1^2 \left(\frac{p_1}{2p_2} x_1 \right) = \frac{p_1}{2p_2} x_1^3 \\ \Rightarrow x_1 &= \left(\frac{2p_2}{p_1} e^{u^*} \right)^{\frac{1}{3}} = \left(\frac{2p_2}{p_1} \right)^{\frac{1}{3}} e^{\frac{u^*}{3}} \end{aligned}$$

And then plugging back into the optimal spending condition,

$$x_2 = \frac{p_1}{2p_2} x_1 = \frac{p_1}{2p_2} \left(\frac{2p_2}{p_1} e^{u^*} \right)^{\frac{1}{3}} = \left(\frac{p_1}{2p_2} \right)^{\frac{2}{3}} e^{\frac{u^*}{3}}$$

And the expenditure function is the cost of this bundle. In the chart I simplified as

$$p_1 \left(\frac{2p_2}{p_1} \right)^{\frac{1}{3}} e^{\frac{u^*}{3}} + p_2 \left(\frac{p_1}{2p_2} \right)^{\frac{2}{3}} e^{\frac{u^*}{3}} = p_1^{\frac{2}{3}} p_2^{\frac{1}{3}} e^{\frac{u^*}{3}} \left(2^{\frac{1}{3}} + \left(\frac{1}{2} \right)^{\frac{2}{3}} \right)$$

4. $u(x_1, x_2) = 2x_1 + x_2$. Here, as in point 4 above, we want to allocate money optimally between goods. Since these are perfect substitutes where good 1 has twice the value, it's optimal to buy good 1 only if good 1 has LESS than twice the cost (i.e. $\frac{p_1}{p_2} < 2$ —this is where good 1 always has more bang-per-buck), and good 2 only if $\frac{p_1}{p_2} > 2$; either is optimal at $\frac{p_1}{p_2} = 2$. But the second constraint is the utility constraint, $2x_1 + x_2 = u^*$.

So start with the case $\frac{p_1}{p_2} < 2$. Then it's optimal to buy good 1 only, so $x_2 = 0$; plugging this into the utility constraint, $2x_1 + x_2 = u^*$, we get $x_1 = \frac{u^*}{2}$. So the compensated demand bundle if $p_1 < 2p_2$ is $(x_1, x_2) = (\frac{u^*}{2}, 0)$ and expenditure function is its cost, $p_1 \frac{u^*}{2} + p_2 \cdot 0$.

Now take the case $\frac{p_1}{p_2} > 2$. Then good 1 always has less bang-per-buck, so it's optimal to buy good 2 only, so $x_1 = 0$. Plugging this into the utility constraint, $2x_1 + x_2 = u^*$, we get $x_2 = u^*$. This gives the compensated demand function in the chart for the range $p_1 > 2p_2$, and the expenditure function here is its cost, $p_1 \cdot 0 + p_2 u^*$.

(For the expenditure function in the chart, I simplified what we have here, $\begin{cases} \frac{p_1}{2} u^* & \text{if } p_1 < 2p_2 \\ p_2 u^* & \text{if } p_1 > 2p_2 \end{cases}$).

5. $u(x_1, x_2) = 2x_1^2 + 2x_2^2$

Here, we know these are concave ($\frac{MU_1}{MU_2}$ RISES as $x_1 \uparrow$ and $x_2 \downarrow$). So it's optimal to pick a corner solution, $x_1 = 0$ or $x_2 = 0$. And the second equation is the utility constraint, $2x_1^2 + 2x_2^2 = u^*$. The best corner is whichever gives utility u^* more cheaply:

- if we set $x_1 = 0$, then the utility constraint is $2 \cdot 0^2 + 2x_2^2 = u^* \Rightarrow x_2 = \sqrt{\frac{u^*}{2}}$: this bundle costs $p_2 \sqrt{\frac{u^*}{2}}$
- if we set $x_2 = 0$, then the utility constraint is $2x_1^2 + 2 \cdot 0^2 = u^* \Rightarrow x_1 = \sqrt{\frac{u^*}{2}}$: this bundle costs $p_1 \sqrt{\frac{u^*}{2}}$
- comparing costs, the first one is best iff $p_2 < p_1$. So the compensated demand bundle is

$$(x_1^c, x_2^c) = \begin{cases} \left(0, \sqrt{\frac{u^*}{2}}\right) & \text{if } p_2 < p_1 \\ \left(\sqrt{\frac{u^*}{2}}, 0\right) & \text{if } p_1 < p_2 \end{cases}$$

(either is optimal at $p_1 = p_2$). And the expenditure function is the cost of this bundle:

$$e(p_1, p_2, u^*) = p_1 x_1^c + p_2 x_2^c = \min\{p_1, p_2\} \cdot \sqrt{\frac{u^*}{2}}$$

4 Income and Substitution Effects

4.1 Question 1.

For Example 2 above, $u(x_1, x_2) = x_1^2 + x_1 x_2$, use the regular and compensated demand functions in the chart above to calculate the income and substitution effects if $p_2 = 1, m = 10$, and p_1 increases from 1 to 3. Also use the expenditure function to calculate the CV.

Answer:

- for x^o , we want the optimal bundle (regular demand) at the original budget: $p_1 = 1, p_2 = 1, m = 10$. Well from the chart on page 1,

$$(x_1, x_2)(p_1, p_2, m) = \begin{cases} \left(\frac{m}{p_1}, 0\right) & \text{if } p_1 < 2p_2 \\ \left(\frac{m}{2(p_1 - p_2)}, \left(\frac{p_1}{p_2} - 2\right) \frac{m}{2(p_1 - p_2)}\right) & \text{if } p_1 > 2p_2 \end{cases}$$

Since here $p_1 < 2p_2$, the top part applies, so $(x_1^o, x_2^o) = \left(\frac{m}{p_1}, 0\right) = (10, 0)$. And thus original utility is $u^o = u(10, 0) = 10^2 + 10 \cdot 0 = 100$

- for x^c , we want the bundle that earns original utility $u^o = 100$ (previous bullet point), but that is optimized at the new prices $p_1 = 3, p_2 = 1$. In other words, the compensated demand function evaluated at $p_1 = 3, p_2 = 1, u^* = 100$. From the compensated demand function chart, we have

$$(x_1^c, x_2^c)(p_1, p_2, u^*) = \begin{cases} (\sqrt{u^*}, 0) & \text{if } p_1 < 2p_2 \\ \left(\sqrt{\frac{p_2 u^*}{p_1 - p_2}}, \left(\frac{p_1 - 2p_2}{p_2} \right) \sqrt{\frac{p_2 u^*}{p_1 - p_2}} \right) & \text{if } p_1 > 2p_2 \end{cases}$$

Here, with $p_1 = 3$ and $p_2 = 1$, we are in the range $p_1 > 2p_2$, so the bottom part of the chart applies. Thus

$$(x_1^c, x_2^c) = \left[\sqrt{\frac{p_2 u^*}{p_1 - p_2}}, \left(\frac{p_1 - 2p_2}{p_2} \right) \sqrt{\frac{p_2 u^*}{p_1 - p_2}} \right]_{p_1=3, p_2=1, u^*=100} = (5\sqrt{2}, 5\sqrt{2})$$

- for CV, we want the subsidy needed to afford (x_1^c, x_2^c) at the new prices. If you do this directly: At new prices $p_1 = 3$ and $p_2 = 1$, it costs $3 \cdot 5\sqrt{2} + 1 \cdot 5\sqrt{2} = \28.284 . You could also get this from the expenditure function: from the compensated demand chart for the range $p_1 > 2p_2$, we have $e(p_1, p_2, u^*) = 2\sqrt{p_2(p_1 - p_2)u^*}$, and thus to get old utility $u^o = 100$ at new prices $p_1 = 3$ and $p_2 = 1$, it costs $\left[2\sqrt{p_2(p_1 - p_2)u^*} \right]_{p_1=3, p_2=1, u^*=100} = \28.284 . Actual income is \$10. So

$$CV = \$28.284 - \$10 = \$18.284$$

- for x^f : we want the final bundle, at new prices $p_1 = 3$ and $p_2 = 1$ but original income $m = 10$. These prices are in the range $p_1 > 2p_2$, so plug into the bottom part of the regular demand expression (first bullet point in this solution). Thus,

$$(x_1^f, x_2^f) = \left[\left(\frac{m}{2(p_1 - p_2)}, \left(\frac{p_1}{p_2} - 2 \right) \frac{m}{2(p_1 - p_2)} \right) \right]_{m=10, p_1=3, p_2=1} = (2.5, 2.5)$$

- finally, substitution effect is $x_1^c - x_1^o = 5\sqrt{2} - 10 = -2.9289$, and income effect is $x_1^f - x_1^c = 2.5 - 5\sqrt{2} = -4.5711$

4.1.1 Question 2

For example 1, $u(x_1, x_2) = \min\{x_1, 2x_2\}$, directly calculate the income and substitution effects if $m = 20$, $p_2 = 2$, and p_1 rises from 1 to 3.

Answer:

- for the original bundle x^o , we want to find the best \$20 bundle at original prices $p_1 = 1$ and $p_2 = 2$. For these perfect complement preferences, we want to avoid wasting money, so equate the expressions in brackets to get $x_1 = 2x_2$. Then plug this into the budget line, $x_1 + 2x_2 = 20$, and solve to get $(x_1^o, x_2^o) = (10, 5)$. And original utility is $u^o = u(10, 5) = \min\{10, 2 \cdot 5\} = 10$.
- for the compensated bundle x^c , we want the bundle that's as good as $x^o = (10, 5)$, but allocates money between goods in the best possible way at the new prices $p_1' = 3$ and $p_2 = 2$. Well, the smartest way to allocate money for these perfect complement preferences is still to equate the expressions in brackets, so $x_1 = 2x_2$. And thus to get utility 10 in the smartest way, in this case we need to stick with the old bundle: it's still $(x_1^c, x_2^c) = (10, 5)$

- for the final bundle x^f , we want the optimal bundle at new prices $p_1 = 3, p_2 = 1$ but actual income $m = 20$. The optimal way to allocate money is again $x_1 = 2x_2$; plug this into the new budget line, $3x_1 + 2x_2 = 20$, and solve to obtain $x_2 = \frac{20}{8}$ and $x_1 = \frac{40}{8}$. So the final bundle is $(x_1^f, x_2^f) = (\frac{40}{8}, \frac{20}{8}) = (5, 2.5)$
- substitution effect is $x_1^c - x_1^o = 10 - 10 = 0$, and income effect is $x_1^f - x_1^c = \frac{40}{8} - 10 = -5$
- CV is the subsidy needed to afford $x^c = (10, 5)$ at new prices $(p'_1, p_2) = (3, 1)$. The cost of the bundle here is $10 \cdot 3 + 5 \cdot 2 = \40 . Actual income is \$20. So the subsidy is \$20. (Notice that here, the substitution effect was zero—there was no way to reoptimize—and so the subsidy needed to maintain utility is the full increased cost of the original bundle: 10 units of good 1 initially cost $10 \cdot 1$ and now cost $10 \cdot 3$, an increase of \$20).

4.2 Question 3

For example 3 above, $\ln x_1^2 + \ln x_2$, find the income and substitution effects if $m = 300, p_2 = 4$, and p_1 increases from 1 to 2. Do this both directly and using the demand and expenditure functions in the charts.

Answer:

- first, find the original bundle. These are equivalent to the Cobb-Douglas preferences $x_1^{\frac{2}{3}} x_2^{\frac{1}{3}}$, so the consumer will optimally buy $x_1 = \frac{2}{3} \frac{m}{p_1} = \frac{2}{3} \frac{300}{1} = 200$ and $x_2 = \frac{1}{3} \frac{m}{p_2} = \frac{1}{3} \frac{300}{4} = 25$. Thus original utility is $u^o = \ln(200^2) + \ln 25 = 13.816$.
- now the compensated bundle. To find it directly, find the bundle that allocates \$ optimally at new prices, so

$$\frac{MU_1}{MU_2} = \frac{p'_1}{p_2} \Rightarrow \frac{2/x_1}{1/x_2} = \frac{2}{4} \Rightarrow x_1 = 4x_2$$

and that earns original utility, so

$$\ln x_1^2 + \ln x_2 = 13.816 \Leftrightarrow \ln(x_1^2 x_2) = 13.816 \Leftrightarrow x_1^2 x_2 = e^{13.816}$$

Plugging in $x_1 = 4x_2$ from the optimality condition,

$$(4x_2)^2 x_2 = e^{13.816} \Rightarrow x_2^3 = \frac{e^{13.816}}{16} \Rightarrow x_2 = \left(\frac{e^{13.816}}{16} \right)^{\frac{1}{3}} = 39.692$$

And then plug this back into $x_1 = 4x_2$ to get $x_1 = 4(39.692) = 158.77$. So $(x_1^c, x_2^c) = (158.77, 39.692)$

To find it using the expressions in the chart, just plug $p_1 = 2, p_2 = 4, u^* = 13.816$ into the compensated demand function:

$$\left[\left(\left(\frac{2p_2}{p_1} \right)^{\frac{1}{3}} e^{\frac{u^*}{3}}, \left(\frac{p_1}{2p_2} \right)^{\frac{2}{3}} e^{\frac{u^*}{3}} \right) \right]_{u^*=13.816, p_1=2, p_2=4} = (158.77, 39.692)$$

- for the CV directly, we want to know how much of a subsidy is needed to afford $x^c = (158.77, 39.692)$ at new prices $(p'_1 = 2, p_2 = 4)$. This bundle costs $2 \cdot 158.77 + 4 \cdot 39.692 = 476.31$. Actual income is \$300. So the subsidy needed is

$$CV = 476.31 - 300 = 176.31$$

If you want to use the expenditure function from the compensated demand chart above, we had

$$e(p_1, p_2, u^*) = p_1^{\frac{2}{3}} p_2^{\frac{1}{3}} e^{\frac{u^*}{3}} \left(2^{\frac{1}{3}} + \left(\frac{1}{2} \right)^{\frac{2}{3}} \right)$$

So if the goal is to get old utility $u^o = 13.816$ at new prices $p'_1 = 2$ and $p_2 = 4$, just plug these in: it now costs

$$\left[p_1^{\frac{2}{3}} p_2^{\frac{1}{3}} e^{\frac{u^*}{3}} \left(2^{\frac{1}{3}} + \left(\frac{1}{2} \right)^{\frac{2}{3}} \right) \right]_{p_1=2, p_2=4, u^*=13.816} = \$476.31$$

an excess of \$176.31 compared to actual income \$300, so this is the subsidy needed, the CV.

- and finally, the final bundle: what do you actually buy at new prices $(p'_1, p_2) = (2, 4)$ but original unsubsidized income \$300. Well again, these are equivalent to Cobb-Douglas preferences $x_1^{\frac{2}{3}} x_2^{\frac{1}{3}}$, so it's optimal to spend $\frac{2}{3}$ of income on good 1, which lets you buy $x_1 = \frac{2}{3} \frac{m}{p_1} = \frac{2}{3} \frac{300}{2} = 100$ units, and $\frac{1}{3}$ of income on good 2, which gets you $x_2 = \frac{1}{3} \frac{m}{p_2} = \frac{1}{3} \frac{300}{4} = 25$ units. So $(x_1^f, x_2^f) = (100, 25)$
- finally, substitution effect is $x_1^c - x_1^o = 158.77 - 200 = -41.23$ and income effect is $x_1^f - x_1^c = 100 - 158.77 = -58.77$. (In words: ultimately, doubling the price of good 1 reduces good 1 consumption by 100 units, from 200 to 100: Specifically, 41.23 units of this are due to reoptimization (even if the consumer got a \$176.31 subsidy, which would be enough to maintain initial utility, they'd still reduce x_1 by 41.23 as they've found a better way to maintain utility at the new prices); and 58.77 units of this are due to income effects (without actually getting a \$176.31 subsidy, they are poorer, and this leads to a 58.77 reduction in good 1 consumption))